

Homework 3

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```
library(astsa)
```

I.

2.10)

We have $x_t = \mu + w_t + \theta w_{t-1}$, $w_t \sim wn(0, \sigma_w^2)$.

a)

$$E[x_t] = E[\mu + w_t + \theta w_{t-1}] = \mu + 0 + 0 = \mu$$

b)

$$\gamma_x(0) = Var(x_t) = Var(\mu + w_t + \theta w_{t-1}) = Var(w_t) + \theta^2 Var(w_{t-1}) = \sigma_w^2(1 + \theta^2)$$

$$\gamma_x(1) = Cov(x_{t+1}, x_t) = Cov(\mu + w_{t+1} + \theta w_t, \mu + w_t + \theta w_{t-1}) = Cov(\theta w_t, w_t) = \sigma_w^2 \theta$$

$$\gamma_x(-1) = Cov(x_{t-1}, x_t) = Cov(\mu + w_{t-1} + \theta w_{t-2}, \mu + w_t + \theta w_{t-1}) = Cov(w_{t-1}, \theta w_{t-1}) = \sigma_w^2 \theta$$

c)

For x_t to be stationary expected value needs to be constant and not depend on t . The Auto-Covariance function needs to depend only on the lag $h = |s - t|$ and not the time s or t . We can see both of these are met in the last two results. For x_t to be stationary for all $\theta \in \mathbb{R}$ we

simply need to show that $\gamma_x(0) = \text{Var}(x)$ is always non negative. Since $(1 + \theta^2)$ is always non negative, we can say that x_t is stationary for all $\theta \in \mathbb{R}$.

d)

We have $\text{Var}(\bar{x}) = \gamma_x(0)/n$ since x_t is white noise.

i)

$$\text{Var}(\bar{x}) = \gamma_x(0)/n = 2\sigma_w^2/n$$

ii)

$$\text{Var}(\bar{x}) = \gamma_x(0)/n = \sigma_w^2/n$$

iii)

$$\text{Var}(\bar{x}) = \gamma_x(0)/n = 2\sigma_w^2/n$$

e)

The accuracy in the estimate of μ is best when $\theta = 0$, the variance of our estimate is 1/2 that of $\theta = 1$ and $\theta = -1$.

2.11)

a)

We are generating the series $x_t = w_t$ where $w_t \sim wn(0, \sigma_w^2)$.

```
set.seed(3)
w = rnorm(500)
acf(w, lag.max = 20, plot = FALSE)
```

Autocorrelations of series 'w', by lag

0	1	2	3	4	5	6	7	8	9	10
1.000	-0.013	-0.025	0.028	0.022	-0.011	0.072	0.016	-0.010	-0.039	-0.007
11	12	13	14	15	16	17	18	19	20	
0.004	-0.057	0.025	-0.106	-0.109	-0.004	-0.013	-0.079	-0.026	-0.003	

We find the actual ACF:

$$\gamma_x(0) = Var(x_t) = \sigma_w^2$$

$$\gamma_x(1) = Cov(x_t, x_{t-1}) = Cov(w_t, w_{t-1}) = 0$$

So:

$$\gamma_x(h) = \begin{cases} \sigma_w^2 & h = 0 \\ 0 & h \geq 1 \end{cases}$$

And then the actual ACF is:

$$\rho(h) = \begin{cases} 1 & h = 0 \\ 0 & h \geq 1 \end{cases}$$

We can see that the sample ACF varies slightly around the true value 0.

b)

```
set.seed(4)
w_2 = rnorm(50)
acf(w_2, lag.max = 20, plot = FALSE)
```

Autocorrelations of series 'w_2', by lag

0	1	2	3	4	5	6	7	8	9	10
1.000	0.003	-0.047	-0.074	0.087	0.133	0.187	-0.025	0.062	-0.217	-0.029
11	12	13	14	15	16	17	18	19	20	
0.255	-0.050	0.041	0.092	0.012	-0.019	0.013	0.081	0.029	-0.076	

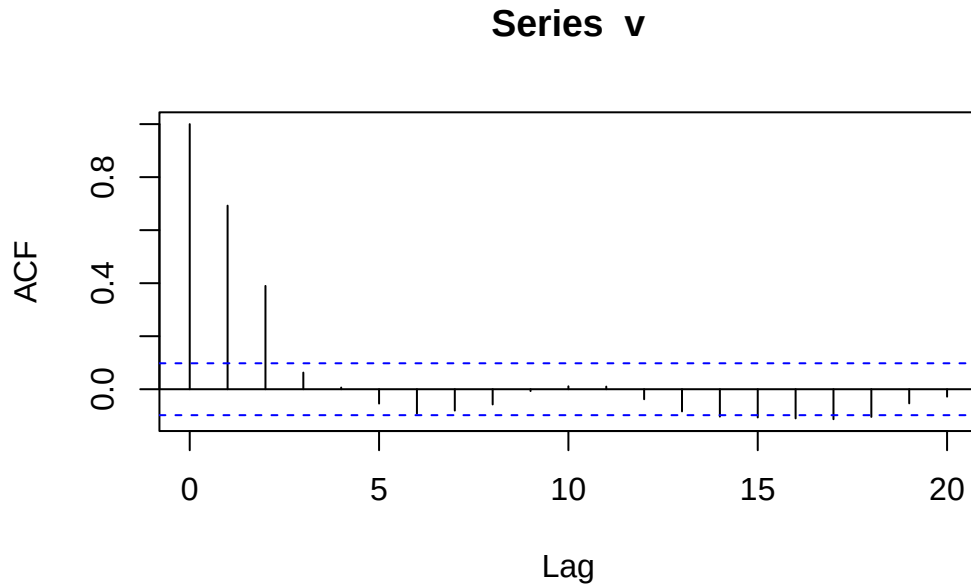
Now the estimates of the ACF vary more than with $n = 500$.

2.12)

We are generating the series $v_t = \frac{1}{3}(w_{t-1} + w_t + w_{t+1})$ where $w_t \sim wn(0, \sigma_w^2)$.

```
set.seed(11)
w = rnorm(500+51)
```

```
v = filter(w, sides = 2, filter = rep(1/3, 3))[100:500]
acf(v, lag.max = 20, plot = TRUE)
```



```
acf(v, lag.max = 20, plot = FALSE)
```

Autocorrelations of series 'v', by lag

0	1	2	3	4	5	6	7	8	9	10
1.000	0.692	0.390	0.063	0.006	-0.054	-0.092	-0.081	-0.057	-0.007	0.011
11	12	13	14	15	16	17	18	19	20	
0.010	-0.038	-0.084	-0.103	-0.106	-0.110	-0.113	-0.104	-0.053	-0.028	

The actual ACF is as follows:

$$\gamma_v(0) = \text{Var}(v_t) = \frac{1}{3}\sigma_w^2$$

$$\gamma_v(1) = \text{Cov}(v_{t+1}, v_t) = \text{Cov}\left(\frac{1}{3}(w_t + w_{t+1} + w_{t+2}), \frac{1}{3}(w_{t-1} + w_t + w_{t+1})\right) = \frac{2}{9}\sigma_w^2$$

$$\gamma_v(2) = Cov(v_{t+2}, v_t) = Cov(\frac{1}{3}(w_{t+1} + w_{t+2} + w_{t+3}), \frac{1}{3}(w_{t-1} + w_t + w_{t+1})) = \frac{1}{9}\sigma_w^2$$

So we have:

$$\gamma_v(h) = \begin{cases} \frac{3}{9}\sigma_w^2 & h = 0 \\ \frac{2}{9}\sigma_w^2 & h = 1 \\ \frac{1}{9}\sigma_w^2 & h = 2 \\ 0 & h \geq 3 \end{cases}$$

And the ACF is as follows:

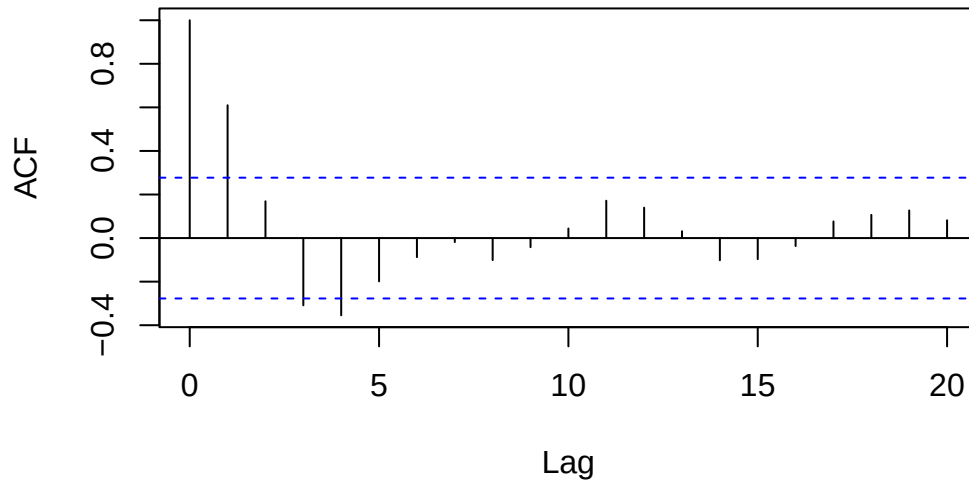
$$\rho_v(h) = \begin{cases} 1 & h = 0 \\ \frac{2}{3} & h = 1 \\ \frac{1}{3} & h = 2 \\ 0 & h \geq 3 \end{cases}$$

The sample ACF is pretty close to the true values.

b)

```
set.seed(15)
w_2 = rnorm(50+51)
v_2 = filter(w, sides = 2, filter = rep(1/3, 3))[2:51]
acf(v_2, lag.max = 20, plot = TRUE)
```

Series v_2



```
acf(v_2, lag.max = 20, plot = FALSE)
```

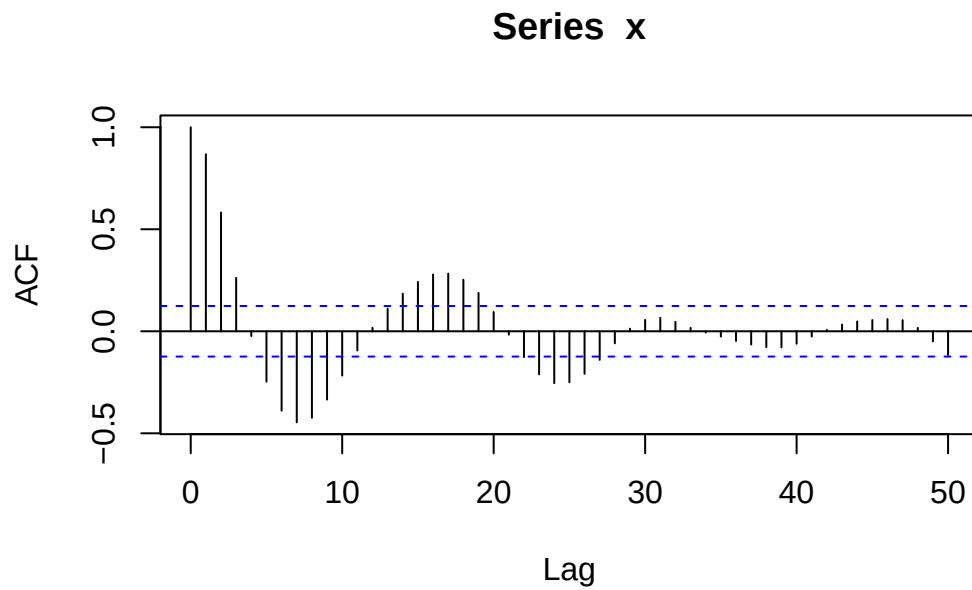
Autocorrelations of series 'v_2', by lag

0	1	2	3	4	5	6	7	8	9	10
1.000	0.610	0.169	-0.309	-0.355	-0.199	-0.088	-0.019	-0.101	-0.042	0.044
11	12	13	14	15	16	17	18	19	20	
0.172	0.140	0.032	-0.102	-0.097	-0.037	0.076	0.107	0.127	0.082	

We can see that the estimates start to vary from the true values by quite a bit, lag 2 is .169 which is quite far from the true $1/3$, and lags 3 and 4 are both far from 0. By reducing n we have increased the variance in our estimator.

2.13)

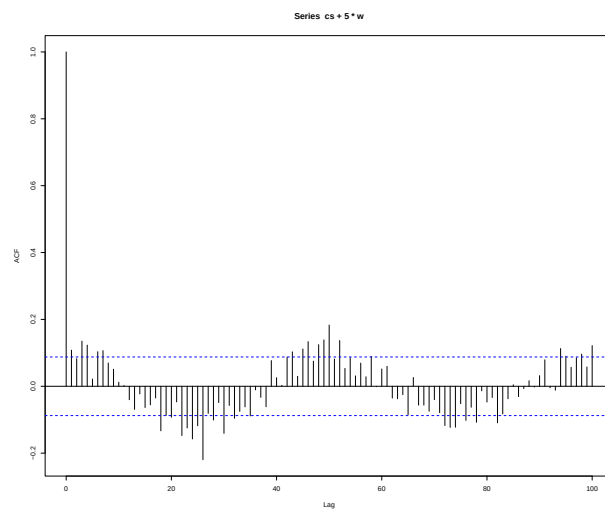
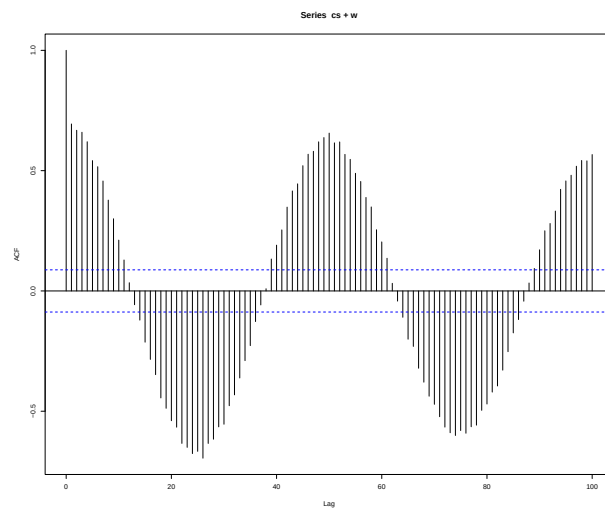
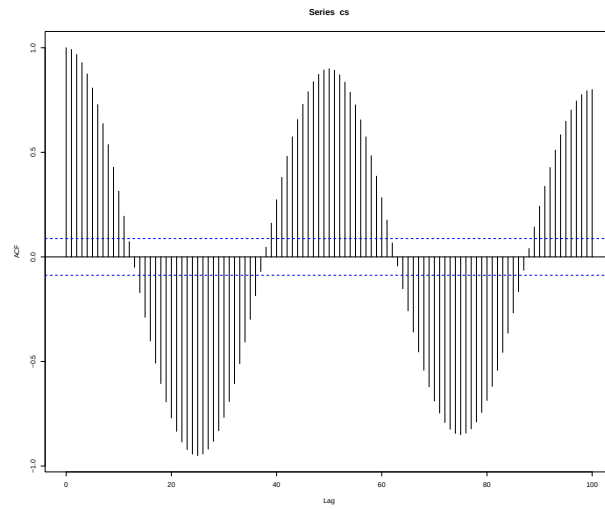
```
set.seed(1000)
w = rnorm(250 + 50)
x = filter(w, filter = c(1.5, -.75), method = "recursive")[-(1:50)]
acf(x, lag.max = 50, plot = TRUE)
```



We can see a strong negative correlated cycle starting at lag 4, that alternates to positive correlation and back to negative correlation about every 8 observations. The structure gradually fades to near zero as observations become more distant.

2.14)

```
t = 1:500
cs = 2*cos(2*pi*(t+15)/50)
w = rnorm(500)
par(mfrow = c(3,1))
acf(cs,lag.max = 100, plot = TRUE)
acf(cs + w,lag.max = 100, plot = TRUE)
acf(cs + 5*w,lag.max = 100, plot = TRUE)
```



We can see that as the white noise component becomes larger the magnitude of the cycle becomes smaller in the ACF.

II.